

1. Declare an integer variable called likes and a pointer variable called ptrLikes; assign likes a value, point ptrLikes to likes, and print both the value and the address stored in ptrLikes.

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int likes = 1000;
```

```
    int *ptrLikes;
```

```
    ptrLikes = &likes;
```

```
    printf("Likes value = %d\n", likes);
```

```
    printf("Address stored in ptrLikes = %p\n", (void *)ptrLikes);
```

```
    return 0;
```

```
}
```

Easy logic:

- likes stores 1000.
- ptrLikes is a pointer.
- ptrLikes = &likes stores the address of likes.
- %p prints an address.

2. Write a function swapPlaylistCounts(int *a, int *b) that swaps the number of songs in two Spotify playlists using pointers, then call the function in main and print the swapped values.

```
#include <stdio.h>
```

```

void swapPlaylistCounts(int *a, int *b)
{
    int temp;

    temp = *a;
    *a = *b;
    *b = temp;
}

int main()
{
    int spotifyPlaylist1 = 50;
    int spotifyPlaylist2 = 100;

    printf("Before Swap:\n");
    printf("Playlist 1 = %d\n", spotifyPlaylist1);
    printf("Playlist 2 = %d\n", spotifyPlaylist2);

    swapPlaylistCounts(&spotifyPlaylist1, &spotifyPlaylist2);

    printf("\nAfter Swap:\n");
    printf("Playlist 1 = %d\n", spotifyPlaylist1);
    printf("Playlist 2 = %d\n", spotifyPlaylist2);

    return 0;
}

```

Easy logic:

& sends the address, and * accesses the value stored at that address.

**3. Given an array of 5 order amounts (e.g., Zomato orders), use a pointer to iterate through the array and print each amount along with its memory address.

Hint: Use pointer arithmetic to move to the next element.**

```
#include <stdio.h>

int main()
{
    int orders[5] = {250, 400, 150, 600, 300};
    int *ptr = orders;

    for (int i = 0; i < 5; i++)
    {
        printf("Order Amount = %d, Address = %p\n",
            *(ptr + i), (void *) (ptr + i));
    }

    return 0;
}
```

Easy logic:

- ptr points to the first element.
- *(ptr + i) gets the value.
- (ptr + i) gives the address of each element.

4. Create a function incrementFollowers(int *followers, int n) that increases each follower count in an array (representing Instagram followers for 5 friends) by 100 using pointer arithmetic, then print the updated counts.

```
#include <stdio.h>
```

```
void incrementFollowers(int *followers, int n)
```

```
{  
    for (int i = 0; i < n; i++)  
    {  
        *(followers + i) = *(followers + i) + 100;  
    }  
}
```

```
int main()
```

```
{  
    int followers[5] = {1000, 2000, 1500, 3000, 2500};
```

```
    incrementFollowers(followers, 5);
```

```
    printf("Updated Followers:\n");
```

```
    for (int i = 0; i < 5; i++)  
    {  
        printf("%d\n", followers[i]);  
    }
```

```
    return 0;
```

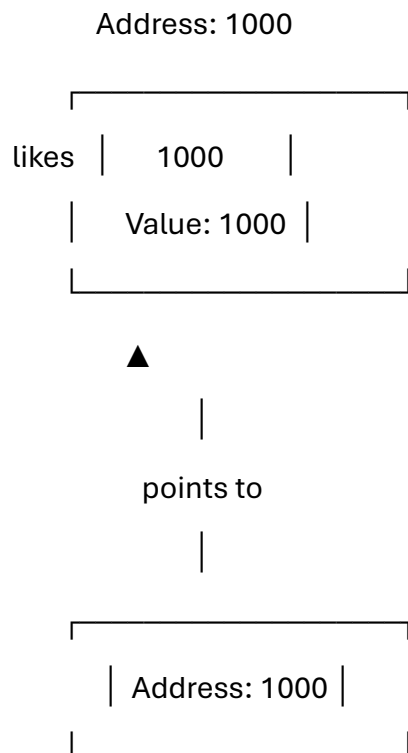
```
}
```

Easy logic:

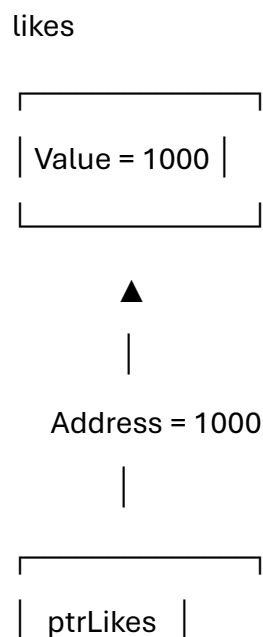
Each follower count is increased by 100 using pointer arithmetic.

5. Draw a simple memory diagram (on paper or using any drawing tool) showing how a pointer references the address of a variable, using the variables from your likes and ptrLikes example; label the variable, pointer, and address.

You can draw this **on paper**:



Or draw it like this:



| stores 1000 |



Pointer